

pharmaniaga®

Morphine Sulphate 10mg/mL Injection



PRODUCT DESCRIPTION

A colourless solution to pale yellow solution.

PHARMACODYNAMICS

Morphine acts as agonists interacting with stereospecific and saturable binding sites or receptors in the brain and other tissues. They are present in highest concentration in the limbic system, thalamus, hypothalamus, midbrain and spinal cord. The affinity of opioid analgesics for the binding sites correlates well with their potency as analgesics.

PHARMACOKINETICS

Absorption: Rapidly absorbed after subcutaneous or intramuscular administration with rapid disappearance from plasma, especially after IV administration.

Blood Concentration: After IV injection of 10 mg, serum concentrations of about 60 ng/mL are obtained in 15 minutes falling to 30 ng/mL after 30 minutes and to 10 ng/mL after 3 hours. After an intramuscular dose of 10 mg, peak serum concentrations of 70 to 80 ng/mL are attained in 10 to 20 minutes; subcutaneous doses given similar concentrations to intramuscular doses at 15 minutes but remain slightly higher during the following 3 hours.

Excretion: After a parenteral dose, about 90% is excreted in 24 hours, with about 10% as free morphine, 65 to 75% as conjugated morphine, 1% as normorphine and 3% as normorphine glucuronide.

Distribution: Widely distributed throughout the body, mainly in the kidneys, liver, lungs and spleen; lower concentrations appear in the brain and muscles; morphine crosses the placenta and traces are secreted in sweat and milk; protein binding, about 35% bound to albumin and to immunoglobulins at concentrations within the therapeutic range.

Metabolic Reactions: Inactivated in the liver, primarily by conjugation with glucuronic acid.

Half-Life: Serum half-life in the period 10 minutes to 6 hours following intravenous administration, 2 to 3

hours; serum half-life in the period 6 hours onwards, 10 to 44 hours.

INDICATION

It is used for symptomatic relief of moderate to severe pain, especially that associated with neoplastic disease, myocardial infarction, and surgery. It is also indicated as adjunctive therapy in the treatment of acute pulmonary oedema secondary to left ventricular failure.

RECOMMENDED DOSAGE

Morphine injections may be given subcutaneously, intramuscularly, intravenously, epidurally, or intrathecally. The duration of use is at the discretion of the physician, and depends on the duration and severity of the disease.

The dosage of Morphine Sulphate 10mg/ml Injection must be adjusted according to the severity of the pain and individual patient response.

The recommended range of individual doses for adults and children is to be understood as a guide for the individually adjusted dosage.

Tabular overview of dosage

Appropriate starting doses are:

Administration	Adults	Children
Intravenous	2.5 - 10 mg	0.05 - 0.1 mg/kg bodyweight
Subcutaneous, intramuscular	5 - 30 mg	0.05 - 0.2 mg/kg bodyweight
Epidural	1 - 4 mg	0.05 - 0.1 mg/kg bodyweight
Intrathecal	0.1 - 1 mg	0.02 mg/kg bodyweight

Adults:

Intramuscular or subcutaneous

5 - 30 mg morphine sulphate. 10 mg is frequently used as a starting dose. This dose may, if necessary, be repeated every 4-6 h.

Intravenous

Only when a particularly rapid onset of effect is necessary. Slowly inject 2.5 – 10 mg morphine sulphate (10 mg per minute, if necessary, dilution with isotonic sodium chloride solution).

Epidural

1 - 4 mg morphine sulphate (diluted with 10 - 15 ml isotonic sodium chloride solution).

Intrathecal

0.1 – 1.0 mg morphine sulphate (diluted with 1 – 4 ml isotonic sodium chloride solution).

Children:

Intramuscular or subcutaneous

0.05 – 0.2 mg morphine sulphate/kg bodyweight;

the single dose should not exceed 15 mg.

Intravenous

Only when a particularly rapid onset of effect is necessary. 0.05 – 0.1 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

Epidural

0.05 – 0.1 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

Intrathecal

0.02 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

The single doses during intramuscular, subcutaneous and intravenous use may be repeated as the effect diminishes, generally every 4 – 6 hours. On account of the longer duration of effect with epidural and especially with intrathecal use, with these methods of administration the daily dose is frequently the same as the individual dose.

Dosage for elderly patients:

Patients of advanced age (generally aged 75 years and above) and patients in poor general physical condition may react more sensitively to morphine. Care must therefore be taken to adjust the dose more carefully and/or choose longer intervals between doses. It may be necessary to switch to lower doses.

A sufficiently high dose must always be given, while at the same time aiming for the lowest analgesically effective dose.

ROUTE OF ADMINISTRATION

Parenteral

[By IM, SC, Epidural, Intrathecal or slow IV injection].

CONTRAINDICATIONS

Respiratory depression, especially in the presence of cyanosis and excessive bronchial secretion, and after operation on biliary tract, acute alcoholism, head injuries and conditions in which intracranial pressure is raised, hypotension, asthma, hepatic impairment, during pregnancy and breast feeding, in patient taking MAO-inhibitors, those with a history of drug abuse. The administration of morphine and other narcotic analgesic are also contraindicated.

WARNINGS AND PRECAUTIONS

Morphine is not usually given preoperatively to children under 1 year of age, and it should be given with extreme care to newborn or premature infants for other conditions. The administration of opioid analgesic during labour may cause respiratory depression in the newborn infant. It should be used with extreme caution in patients with decreased respiratory reserve. It should be used with caution in patients with myasthenia gravis. Alarming and atypical effects sometimes occur in the elderly and the dosage should be reduced in

elderly and debilitated patient.

It should be given with caution or in reduced doses to patients with hypothyroidism, adrenocortical insufficiency, impaired kidney or liver function, prostatic hypertrophy, shock or inflammatory or obstructive bowel disorders

Risk from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of morphine sulphate with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when morphine sulphate is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially lifethreatening condition, have been reported during concurrent use of morphine sulphate with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant

use, buy may occur later than that. Discontinue morphine sulphate if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience).

INTERACTIONS WITH OTHER MEDICAMENTS

Serious and sometimes fatal reaction have occurred when given to patients already receiving monoamine oxidase inhibitors. Depressant effects of morphine are enhanced by CNS depressants such as alcohol, anaesthetics, hypnotics and sedatives, tricyclic antidepressants and phenothiazines. Opioid analgesics with some antagonist activity such as buprenorphine, butorphenol, nalbuphine or pentazocine may precipitate withdrawal symptoms in patients who have recently used morphine. The actions of opioid may in turn affect the activities of other compounds e.g. their gastro-intestinal effects may delay absorption as with maxilefine or may be counter-active as with metoclopramide.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system the control respiration.

Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are

inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue pethidine hydrochloride if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Incompatibilities

Morphine sulphate is physically incompatible with acyclovir sodium, aminophylline, amobarbital sodium, cefepime hydrochloride, chlorothiazide sodium, flucloxacillin sodium, furosemide, gallium nitrate, heparin sodium, meperidine hydrochloride, meperidine sodium, methicillin sodium, minocycline hydrochloride, pentobarbital sodium, phenobarbital sodium, phenytoin sodium, sargramostim, sodium bicarbonate, thiopental sodium.

PREGNANCY AND LACTATION

Pregnancy

An association with increased frequency of inguinal hernias in infants has been postulated in man. Pregnant patients should only be given morphine when the benefits clearly outweigh potential risks to the foetus.

Breast-feeding

Morphine is excreted in human milk and breastfeeding is not recommended while a patient is receiving morphine. Withdrawal syndrome was observed in breastfed infants after maternal administration of morphine sulfate had been stopped.

Use in labour/delivery

Morphine may prolong labour by temporarily reducing the strength, duration and frequency of uterine contractions. Conversely, it may tend to shorten labour by increasing the rate of cervical dilatation. Infants born to mothers receiving opioid analgesics during labour should be observed closely for signs of respiratory depression. In such infants a specific opioid antagonist, naloxone hydrochloride, should be available for reversal of narcotic induced respiratory depression. After chronic morphine use by the mother, newborns may develop withdrawal symptoms.

SIDE EFFECTS

Nausea and vomiting, constipation and drowsiness;

larger doses produce respiratory depression and hypotension; other include difficulty with micturition, ureteric or biliary spasm, dry mouth, sweating, facial flushing, vertigo, bradycardia, palpitations, postural hypotension, hypothermia, hallucinations, mood changes, dependence, miosis, urticaria and pruritus.

Post marketing Experience

Serotonin syndrome (See Warnings and Precautions) Adrenal insufficiency (See Warnings and Precautions) Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitarygonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

OVERDOSE AND TREATMENT

Symptoms:

Overdosage with morphine is characterised by respiratory depression (a decrease in respiratory rate and/or tidal volume, Cheyne-Stokes respiration, cyanosis), pinpoint pupils, extreme somnolence progressing to stupor and coma, skeletal muscle flaccidity, cold and clammy skin and sometimes bradycardia and hypotension. In severe overdosage, apnoea, circulatory collapse, cardiac arrest and death may occur.

Treatment:

Immediate attention should be given to the reestablishment of adequate respiratory exchange through provision of a patent airway and institution of assisted or controlled ventilation. Oxygen, intravenous fluid, vasopressors and other supportive measures should be employed as indicated.

The narcotic antagonist, naloxone, is a specific antidote for morphine. The recommended adult dose of naloxone is 0.4 to 2.0 mg IV every 2 to 3 minutes as necessary, simultaneously with assisted respiration. For children, the initial recommended dose is 0.01 mg/kg naloxone. A response should be seen after 2-3 doses. Note that the duration of action of naloxone is usually shorter than that of morphine and thus patients should be carefully monitored for signs of CNS depression returning.

If the response to naloxone is suboptimal or not sustained, additional naloxone may be administered as needed to maintain alertness and respiratory function. There is no additional information available about the cumulative dose of naloxone that may be safely

administered.

To sustain opioid antagonism, an intravenous infusion of naloxone has been suggested. Naloxone (4 jig per ml) in sodium chloride injection (0.9 %) or glucose injection (5 %) may be infused at a rate titrated in accordance with the patient's response both to the infusion and previous bolus injections.

Naloxone should not be administered in the absence of clinically significant respiratory or circulatory depression secondary to morphine overdosage. It should be administered cautiously to persons who are known or suspected to be physically dependent to morphine. In such cases, an abrupt or complete reversal of opioid effects may precipitate an acute withdrawal syndrome. The severity of the withdrawal syndrome will depend on the degree of physical dependence and the dose of the antagonist administered. If it is necessary to treat serious respiratory depression in the physically dependent patient, the antagonist should be administered with extreme care and by titration with smaller than usual doses of the antagonist.

Morphine toxicity may be the result of overdosage but because of the large inter-individual variation in sensitivity to opioids it is difficult to assess the exact dose of any opioid that is toxic or lethal. The toxic effects of morphine tend to be overshadowed by the presence of pain or tolerance. Patients receiving chronic morphine therapy have been known to take in excess of 3000 mg/day with no apparent toxic effect.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Morphine may impair the mental and/or physical abilities required for the performance of potentially hazardous tasks, such as driving a car or operating machinery. Morphine in combination with other narcotic analgesics, phenothiazines, sedative hypnotics, and alcohol have additive depressant effects.

Patients should not drive or operate machinery.

STORAGE CONDITIONS

Store below 30°C. Protect from light.

Do not use if solution contains growth, particles, turbidity or if its color is darker than pale yellow.

SHELF LIFE

3 years.

DOSAGE FORM AND PACKAGING AVAILABLE

• 10 x 1mL ampoule (amber).

PRODUCT REGISTRATION HOLDER/ MANUFACTURER

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